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How AI social responsibility shapes consumers' brand advocacy

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This paper employs a mixed-methods research approach to investigate the dimensional structure of AI social responsibility and its influence mechanism on consumer brand advocacy. First, based on in-depth interviews with 20 Chinese consumers (conducted January–February 2024) and grounded theory analysis, AI social responsibility(AISR) was classified into two dimensions: technical AISR and institutional AISR. Exploratory factor analysis, confirmatory factor analysis, and structural validity tests were further applied to questionnaire data collected from an online survey platform (Credamo) in 2024 to validate the rationality of these dimensions and their scales. Second, this paper found through two experimental studies that institutional AISR is more likely to inspire consumers' brand advocacy than technical AISR. Moreover, perceived AI credibility and perceived AI empathy mediate the impact of AI social responsibility on consumers' brand advocacy, where AI literacy positively moderates these effects. This paper not only enriches a two-dimensional conceptualization (Technical vs. Institutional AISR) and a validated measurement scale for AISR in the consumer context but also reveals the dual mediating pathways (perceived AI credibility for Technical AISR and perceived AI empathy for Institutional AISR) through which distinct AISR dimensions differentially drive consumer brand advocacy. Furthermore, it uncovers the critical moderating role of AI literacy in amplifying these effects. These findings extend prior CSR and AI-related consumer behavior literature by delineating the nuanced mechanisms and boundary conditions of AISR's impact.

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Introduction

As artificial intelligence (AI) becomes deeply embedded in business operations and consumer interactions worldwide, the imperative for Artificial Intelligence Social Responsibility (AISR) has surged to the forefront of academic discourse and corporate strategy. This global concern is underscored by international frameworks, such as the UNESCO Recommendation on the Ethics of AI and the EU AI Act, which emphasize the importance of fairness, transparency, and accountability. High-profile incidents, such as algorithmic bias in hiring tools disadvantaging certain demographic groups across different regions (Akter et al. 2023), large-scale data breaches involving AI systems impacting millions globally (Turjeman and Feinberg 2024), and debates over the opacity of AI decision-making ('black box' problem) raising accountability concerns, highlight the pervasive ethical and social challenges that demand corporate responsibility. Leading technology firms, including Microsoft, Google, and IBM, have publicly established AI ethics principles and governance structures in response, acknowledging the critical link between responsible AI practices and sustained consumer trust and brand value. As AISR gains prominence in shaping brand values and consumer-brand interactions globally (Manser Payne et al. 2021), the ethical challenges posed by AI can elicit complex consumer emotions (Du and Xie 2021). From a social psychological perspective, advanced technology enhances brand appeal (Liu and Lei 2025). However, if consumers encounter negative experiences while engaging with AI services, it can significantly undermine their trust in AI products, posing a potential threat to the company's brand image (Jain et al. 2024). Thus, a nuanced understanding of the nexus between AI social responsibility(AISR) and consumers' brand advocacy is imperative for corporate strategy.

The proliferation of AI technology catalyzes relentless innovation within the business sphere, yet it concurrently prompts introspection on the societal responsibilities enterprises ought to embrace (Dignum, V. 2020). Despite its significance, the potential of AI Social Responsibility (AISR) in brand relationship management remains an uncharted territory in academic discourse.

While the burgeoning literature on responsible AI, algorithmic responsibility, and AI ethics (Wang et al. 2020; Xiao 2022; Huang et al. 2023) provides valuable insights into the ethical imperatives of AI, and CSR research establishes a foundation for understanding corporate citizenship (Godfrey et al. 2009), gaps persist when it comes to understanding AISR specifically within the consumer domain and its impact on crucial consumer outcomes like brand advocacy. First, there is a dearth of multidimensional conceptual frameworks for AISR tailored to the consumer context. Existing work tends to focus either on technical aspects (e.g., privacy breaches, algorithmic bias—Turjeman and Feinberg 2024; Akter et al. 2023), on ethical norms articulated at a high level of abstraction (e.g., AI ethics principles), or on broader societal impacts (digital addiction, fairness—Du and Sen 2023). Second, stemming from the conceptual gap, there is a critical absence of validated measurement instruments for AISR. Research on responsible AI often relies on proxies or measures designed for specific, narrow issues (e.g., Kumar et al. 2023 used scales adapted for healthcare AI responsibility), lacking a scale that captures the full spectrum of consumer-relevant AISR dimensions. This methodological gap impedes empirical investigation into the antecedents and consequences of AISR. Third, the mechanisms linking AISR to deep consumer engagement, specifically brand advocacy, are poorly understood. While CSR research suggests a positive link to brand relationships (Xie et al. 2019), and AI studies touch on trust or satisfaction (Wirtz and Lwin 2009; Zein 2021), there is scant research that: (a) explicitly examines how distinct dimensions of AISR (e.g., technical vs. institutional) differentially influence consumer brand advocacy; (b) uncovers the mediating psychological processes (such as perceived AI credibility and empathy) through which these effects occur; and (c) identifies key moderators (AI literacy) that shape these pathways. This lack of mechanistic insight limits theoretical development and practical guidance for leveraging AISR to foster brand champions.

To address these critical gaps, this research:

1. Develops and validates a multidimensional framework of AISR specifically for the consumer market, identifying its core dimensions (Technical and Institutional AISR) and sub-factors through qualitative and quantitative methods.
2. Constructs and psychometrically validates a reliable scale to measure consumer perceptions of AISR, providing a necessary tool for future research.
3. Theorizes and empirically tests a comprehensive model elucidating how different AISR dimensions influence consumer brand advocacy, specifically highlighting the distinct mediating roles of perceived AI credibility (for Technical AISR) and perceived AI empathy (for Institutional AISR), and the moderating role of AI literacy in amplifying these effects (as conceptualized in Fig. 3). This provides a deeper, more nuanced understanding than prior monolithic approaches to CSR/AI responsibility.

By tackling these gaps, this study aims to advance the theoretical understanding of AISR in consumer behavior and provide actionable insights for businesses navigating the ethical deployment of AI.

The connotation of AI social responsibility

AISR is an evolutionary stage of Corporate Social Responsibility (CSR). CSR has long been anchored in foundational frameworks, such as Carroll's (1991) Pyramid—encompassing economic, legal, ethical, and philanthropic responsibilities—and stakeholder theory, which emphasizes corporate obligations to diverse groups affected by business operations (Parmar et al. 2010). While these models remain pivotal, the advent of AI necessitates a recontextualization of CSR dimensions. Technical AISR, focusing on functional safeguards (e.g., privacy, bias mitigation), aligns with Carroll's economic/legal layers and primary stakeholder needs (e.g., consumers, employees). Institutional AISR, which addresses societal ethics and well-being, extends to the ethical/philanthropic layers, as well as secondary stakeholders (e.g., communities, regulators). Our two-dimensional AISR framework thus adapts these classical lenses to AI-specific challenges. Additionally, D'Cruz et al. (2022) view AI social responsibility as CSR stage 3.0, conceptualizing CSR as including a technological component in its conceptualization - technological social responsibility. They emphasized AISR as a CSR strategy and practice that addresses key ethical and socio-technical issues related to AI technologies on the one hand, and leverages the power of AI and related technologies to address social and environmental problems on the other. There is no consensus among academics on the conceptual naming and connotation interpretation of AI social responsibility, and the related concepts mainly involve three aspects: responsible AI, algorithmic responsibility, and AI ethics. Among them, Wang et al. (2020) proposed the idea of "responsible AI", advocating for the integration of ethical norms and responsibility in the planning and implementation of AI to ensure its auditability and accountability mechanisms, as well as to guarantee that AI behaviors are in line with relevant regulations and standards. Algorithmic responsibility entails effectively managing the impact of algorithmic decisions on society, stakeholders, and the natural

environment through transparent and ethical algorithms, thereby achieving optimal performance in enhancing social welfare (Xiao 2022). Therefore, as algorithms underlie artificial intelligence, their responsibility becomes a focal point in the application of AI technology and should be a concern. In addition, Huang et al. (2023) define AI ethics as a set of moral norms and social values that must be followed during the development and application of AI to thoroughly study and regulate the ethical impacts that may be brought about by AI technology and to ensure that it does not harm society.

In summary, although existing studies do not directly address the conceptual naming and explanation of AI social responsibility, the literature on responsible AI, algorithmic responsibility, and AI ethics provides a reference for this paper to construct further and explain the conceptual connotation of AI social responsibility. Therefore, this paper defines Artificial Intelligence Social Responsibility (AISR) as the obligation of enterprises to strictly adhere to laws and ethics, protect privacy, and enhance accountability and transparency throughout the entire lifecycle of AI technology development and application. The ultimate aim is to maximize stakeholder benefits while promoting human well-being, fairness, and justice.

Dimensions and scale development for AISR

Research procedures and data collection. We invited one AI marketing expert and three PhD students in business administration to participate in designing the interview outline. Through pre-interviews with four respondents, we designed three open-ended questions and corresponding switching/pursuing questions. The study selected consumers who had experienced AI services. During the interviews, we made every effort to ask the interviewees to cite examples to discuss and to ask questions appropriately in conjunction with specific queries. Between January and February 2024, we conducted in-depth interviews with 20 participants with prior experience using artificial intelligence. By the 16th interview, no new themes emerged. According to Glaser et al. (2017), theoretical saturation is achieved in data collection when additional data fails to generate new theoretical categories or concepts. The total interview time was approximately 9 hours, and approximately 60,000 characters of textual information were compiled from the interview data. Refer to Appendix 5 for details on the respondents.

Grounded theory analysis. While prior research on Responsible AI (Wang et al. 2020; Huang et al. 2023)and technology ethics (Hagendorff 2020) offers valuable conceptual foundations, these works predominantly focus on high-level principles (e.g., fairness, transparency), sector-specific applications (e.g., healthcare AI in Kumar et al. 2023), and technical or ethical dimensions in isolation. This paper adopts grounded theory for three reasons: (1) Capturing consumer-centric nuances—existing frameworks lack insights into how consumers perceive and prioritize AISR dimensions in daily interactions (e.g., data traceability vs. emotional value). Inductive coding of interview data (60,000 characters from 20 users) revealed context-specific constructs, such as reflexivity and well-being, which are absent in top-down models. (2) Integrating technical-institutional duality: Prior scales (e.g., Kumar et al.'s 4-item healthcare AI scale) fail to reconcile functional safeguards (technical AISR) with socio-ethical commitments (institutional AISR). Grounded theory enabled us to derive an integrated framework in which safety/usability co-evolve with ethics/well-being. (3) Validating latent constructs: Heuristic constructs like algorithmic manipulation (CRG15-1) and AI-facilitated emotional value (WY5-5) emerged organically from user narratives, demonstrating grounded theory's advantage in uncovering implicit concerns. The flowchart of grounded theory analysis is shown in Fig. 1.

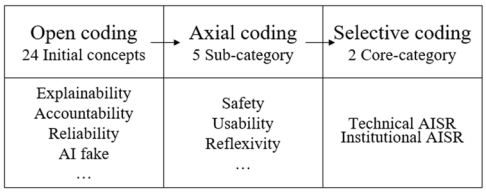

Fig. 1 The flowchart of grounded theory analysis.

Open coding. This study employed a combination of personal coding and expert consultation to meticulously analyze the primary sources, ensuring coding remained as close as possible to the original data. Three coders utilized NVivo 11 software to conduct line-by-line, sentence-by-sentence, and word-by-word coding, ensuring no key phrases were overlooked. They employed original discourse "labeling" during the coding process to accurately reproduce the interviewees' statements, thereby minimizing the coders' subjective bias. After several rounds of comparative analysis, abstract reorganization, and the elimination of invalid concepts, 24 initial concepts were collated. An example of an open coding section is provided in Appendix 1.

Axial coding. The coding team iteratively summarized and organized the 24 concepts obtained from open coding, integrating them with existing research on AISR to ensure that the subcategories accurately reflected the theme of AISR. After iterative discussions, five subcategories related to AISR were refined: safety, usability, reflexivity, ethics, and well-being. Safety pertains to AI's capacity to protect user privacy and guarantee transparent, explainable decision-making processes, encompassing data access, prevention of discriminatory pricing based on big data, data traceability, explainability, and accountability. Usability, on the other hand, is a multifaceted construct that reflects the degree to which a product enables users to achieve specific objectives within defined contexts, characterized by its effectiveness, efficiency, and the satisfaction it delivers to users (Alabduljabbar 2024). This encompasses a spectrum of attributes, including reliability, authenticity, repeated recommendations, labeling, fair access, and algorithmic discrimination. Reflexivity refers to AI's capacity for self-reflection, adaptation, and optimization in response to challenges such as digital addiction, the attribution of responsibility, and information cocooning. For instance, it involves preventing overindulgence and enhancing user autonomy. Ethics encompasses a set of codes and guidelines that ensure AI adheres to societal moral standards, laws, and regulations, while also promoting environmental protection and social progress (Hagendorff 2020). This includes compliance with laws and regulations, alignment with ethical codes, sustainability, prevention of algorithmic manipulation, protection against significant data breaches, avoidance of AI-generated fakes, and addressing AI-related threats. Well-being is the experience of value that consumers have in all aspects of their work and life (El Hedhli et al. 2013), encompassing both the enhancement of the quality of life and work for consumers. To assess inter-rater reliability, we calculated Fleiss's kappa (a measure of inter-rater reliability among three or more raters), which was 0.878 ($p < 0.001$). A kappa value greater than 0.8 indicates that different raters assessed the coding with high similarity (Kindermann et al. 2024). Consequently, the coding results of this study were considered credible.

Selective coding. After extracting the core categories through selective coding, further analysis of their logical relationships is required; that is, the core categories need to be integrated into the

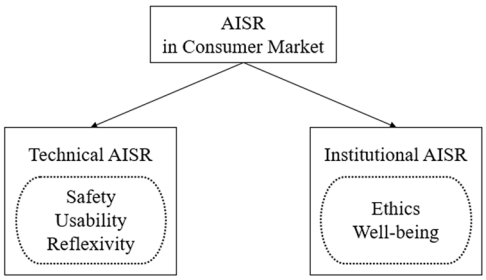

Fig. 2 AISR dimensions and factors.

final theoretical framework. In previous studies, Mattingly and Berman (2006) proposed a dichotomous categorization of technical CSR and institutional CSR based on differences in corporate stakeholders. Technical CSR primarily meets the needs of a company's primary stakeholders, with actions closely linked to the company's profitability goals and demonstrating potential competitive advantages. Institutional CSR focuses more on the company's social responsibility to secondary stakeholders and is manifested as voluntary social welfare practices. AISR involves multiple stakeholders, including technicians (scientists, algorithm engineers), businesspeople (manufacturers, operators, sellers), consumers, social sector individuals (NGOs, media), academics, government officials, and the public (Xue and Pang 2022). When compared with the original interview data, the stakeholders involved in AISR are encompassed within five subcategories. This paper then further categorizes these five subcategories into two main categories: technical AISR and institutional AISR, based on the needs of primary and secondary stakeholders. Technical AISR mainly satisfies consumers' individual functional demands by optimizing algorithmic technology and improving AI product experiences, thereby enhancing the potential competitive advantage of enterprises. It meets primary stakeholder demands and demonstrates potential competitive advantages, such as protecting customer privacy and eliminating algorithmic bias. Institutional AISR primarily satisfies the public's affective demands by embedding ethical and moral values, protecting public AI consumption safety, and thus establishing a good public image. It essentially meets secondary stakeholder demands and represents a voluntary social welfare practice, such as strengthening the ethical design of AI products and reducing public cybersecurity risks.

In summary, based on the grounded theory, this study ultimately achieved a two-dimensional, five-factor conceptualization of AISR, as shown in Fig. 2.

Initial scale generation. First, we derived 24 initial AISR measurement items from the coding results of the aforementioned grounded theory analysis. Second, we evaluated the meaning and precision of each item based on the principles of semantic clarity and conciseness, and eliminated 2 items with repetitive semantics, ambiguity, and poor relevance, such as 'This AI service is authentic' and 'This AI can provide me with emotional value.' Finally, 2 marketing experts and 3 PhD students in the field of AI were invited to review the remaining 22 items. They further deleted 2 items that had the same content but different expressions, or that were ambiguous and unclear, for example, 'The AI recommended the same price of products for different groups of people' and 'The AI did not repeatedly recommend information to me. Consequently, the initial AISR scale, consisting of 20 items, was established.

Exploratory factor analysis (EFA). Between February and March 2024, we distributed 340 questionnaires on Credamo, a well-known online research panel platform in China commonly used for academic studies in the region. After excluding 40 invalid responses (i.e., those failing attention-check items, exhibiting completion times shorter than half the median, or lacking prior experience with AI technology. The data screening in the following text also adopts these criteria), we retained 300 valid questionnaires, yielding an effective response rate of 88%. Credamo's extensive participant pool allows us to reach diverse consumer demographics relevant to our research context conveniently. Among the respondents, 63% were women, 46% were aged 27-35, 77% held undergraduate degrees, and 58% had used AI for 3–5 years. The research utilized a 20-item scale measured on a Likert scale with 7 points.

The results of the statistical analysis revealed that the KMO value was 0.918, and Bartlett's test of sphericity yielded a $\chi 2$ value of 5393.639 with $p < 0.001$. These results indicate that the data sampling was adequate and that there was a strong correlation among the questionnaire items, making the data suitable for factor analysis. Moreover, factor analysis was conducted using principal component analysis with orthogonal rotation, and the cumulative variance contribution of the five factors was 77.975%. Factors with eigenvalues greater than 1 were retained, and items with factor loadings less than 0.6 were excluded. Additionally, the 20 items were further refined based on factor loadings, dimensional alignment, and item similarity, ultimately retaining 18 items with factor loadings ranging from 0.692 to 0.856, indicating good results for each item's factor loadings. Furthermore, the Cronbach's alpha coefficients for the factors were as follows: 'safety' at 0.902 with 4 items; 'usability' at 0.948 with 4 items; 'reflexivity' at 0.884 with 3 items; 'ethics' at 0.884 with 3 items; and 'well-being' at 0.766 with 3 items. The overall reliability of the AISR scale, comprising 18 items, was 0.942, with technical AISR reliability at 0.929 and institutional AISR reliability at 0.877. Thus, the five factors of the AISR demonstrate good explanatory power. Table 1.

Confirmatory factor analysis (CFA). We distributed 431 questionnaires via the Credamo platform(conducted March–April 2024), excluded 28 invalid ones, and retained 403 valid questionnaires, resulting in a validity rate of 93.5%. The research utilized an 18-item scale measured on a Likert scale with 7 points. Initially, the data were tested for reliability. The results indicated that the Cronbach's alpha coefficient for the entire scale was 0.943, and the Cronbach's alpha coefficients for the five factors were 0.858, 0.919, 0.908, 0.919, and 0.895, respectively, indicating good internal consistency. Concurrently, the results of the second-order five-factor model demonstrated that $\chi 2/df = 2.809 < 3$, $CFI = 0.959$, $NFI = 0.939 > 0.90$, $IFI = 0.960 > 0.9$, $GFI = 0.915 > 0.9$, and $RMSEA = 0.067 < 0.08$, suggesting that the model fits the data well. Additionally, the factor loadings for all AISR items were greater than 0.7 and statistically significant. The fit indices of the other factor models are shown in Table 2.

Structural validity test. In this study, the structural validity of AISR was assessed by evaluating reliability (composite reliability, CR), correlation coefficients, and average variance extracted (AVE) for both convergent and discriminant validity, as presented in Table 3. The composite reliability of the five AISR factors ranged from 0.838 to 0.932, all exceeding the benchmark of 0.7, indicating good convergent validity. The AVE values ranged from 0.584 to 0.655, all of which were greater than the benchmark of 0.5. The correlation coefficients between the factors ranged from 0.494 to 0.620, all of which were lower than the square root of their respective AVE values. This suggests a moderate to low correlation between the factors, indicating that

Table 1 Exploratory factor analysis.

Dimensions	Factors	Items	Factor loadings				
Technical	Safety	The data collected by AI told me.	0.820				
		The AI will not illegally use and disclose my personal information.	0.761				
		The information recommended by the AI can be traced back.	0.853				
		The information recommended by the AI is explanatory.	0.709				
	Usability	The AI is reliable in algorithm technology when recommending products.		0.871			
		The information recommended by the AI is not discriminatory.		0.872			
		The AI can meet my functional demands.		0.773			
		The AI service is reliable.		0.805			
	Reflexivity	The AI will remind me that I should not overuse digital technology or services.			0.718		
		The AI allows me to choose my own service or product.			0.789		
		The AI will prompt me to use AI technology healthily.			0.804		
Institutional	Ethics	The AI emphasizes the public consumption ethics.				0.737	
		The AI can comply with the corresponding laws and regulations.				0.751	
		The AI can promote social progress.				0.763	
		The AI will not deceive consumers.				0.756	
	Well-being	The AI satisfies the public's emotional value.					0.791
		The AI improves the quality of life of consumers.					0.708
		The AI improves the quality of work of consumers.					0.795
Reliability coefficient			0.902	0.948	0.884	0.898	0.766
Eigenvalue			9.598	1.578	1.409	1.110	1.063
Cumulative Variance Contribution Value (%)			18.501	36.931	53.778	67.425	79.482

Table 2 Table of overall fit coefficients ($N = 403$).

model	χ^2	df	χ^2/df	IFI	CFI	NFI	GFI	RMSEA
second-order five-factor model	393.287	140	2.809	0.960	0.959	0.939	0.915	0.067
baseline model	664.064	142	4.677	0.917	0.916	0.897	0.839	0.096
four-factor Model(safety+ Usability)	1152.733	146	7.891	0.840	0.839	0.821	0.728	0.131
three-factor Model(safety+ Usability+ Reflexivity)	1582.487	149	10.620	0.772	0.771	0.754	0.656	0.155
Two-factor model (safety+ Usability+ Reflexivity+ Ethics)	1984.597	151	13.143	0.708	0.707	0.691	0.609	0.174
Single-factor model	2672.145	152	17.580	0.598	0.597	0.584	0.557	0.203

Table 3 Correlation coefficients, AVE values, and CR values for each factor ($N = 318$).

Factors	Safety	Usability	Reflectivity	Ethics	Well-being
safety	0.765				
usability	0.502**	0.787			
reflectivity	0.543**	0.620**	0.797		
ethics	0.494**	0.567**	0.592**	0.7518	
well-being	0.494**	0.605**	0.611**	0.569**	0.809
AVE	0.5845	0.6197	0.6359	0.5652	0.6555
CR	0.8487	0.8665	0.8396	0.8387	0.8509

** denotes $P < 0.01$, the diagonal value is the positive square root of AVE

while the various elements of AISR are related to some extent, they also exhibit good discriminant validity, thus forming a cohesive construct.

Hypothesis and research modeling

AI social responsibility and consumers' brand advocacy. Brand advocacy refers to consumers' active promotion and defense of an organization, product, or brand to others, representing the highest level of recognition in the consumer-brand relationship (Walz and Celuch 2010). It is a pivotal concept that reflects the depth of this relationship. Brand advocacy transcends mere communication and dissemination; it includes personal recommendations, attracting potential clients, and staunchly supporting the brand amidst adversity or criticism (Melancon et al. 2011). Brand advocacy extends beyond simple brand recommendations and word-of-mouth information dissemination (Ali et al. 2021); it is a more profound form of active participation that requires consumers to invest significant effort in promoting and defending the brand (Jillapalli and Wilcox 2010).

While CSR plays a significant role in building a positive public image for companies and fostering trusting relationships with consumers (Wang et al. 2023), it is equally important to consider the potential negative impacts of companies' AI technology implementations on the sustainability of their business. These impacts include heightened cybersecurity risks, increased consumer vulnerability to data breaches, and the challenges associated with measuring the effectiveness of technology strategies (Nishant et al. 2020). AI dynamically transforms the brand landscape through creative breakthroughs, innovative content, and immersive experiences (Kar and Varsha 2023). Additionally, the quality of AI services plays a crucial role in shaping the relationship between brands and consumers (Deryl et al. 2023). Therefore, companies should actively engage in AISR to enhance consumer brand advocacy. The core mechanism by which CSR influences consumer brand advocacy is how consumers interpret corporate social responsibility (Xie et al. 2019). Consumers are more likely to engage actively in brand advocacy activities if they view CSR behaviors as positive and proactive (Aljarah et al. 2022).

The impact of CSR behaviors on brand advocacy depends on consumers' perceptions of a company's motives for engaging in

social responsibility. Technical AISR creates competitive advantages for the company through its key stakeholders, while institutional AISR is more widely regarded as a social good practice (Godfrey et al. 2009). Actions taken by companies for key stakeholders, namely technical AISR, may be seen by the public as a basic obligation, essentially for building the company's competitive advantage. This can lead to negative consumer attitudes towards the company (Mantovani et al. 2017), which in turn reduces consumer brand advocacy behavior. Conversely, when consumers witness institutional AISR actions, these actions instill a positive emotional response, leading to spontaneous engagement in brand advocacy (Aljarah et al. 2022). Therefore, from the perspective of AISR categorization, we propose that institutional AISR is more effective in enhancing consumer brand advocacy than technical AISR. In summary, the following hypotheses are proposed:

H1: Institutional AISR is more likely to stimulate consumer brand advocacy than technical AISR.

The mediating role of perceived AI credibility. According to source credibility theory (Hovland and Weiss 1951), the decision-making of information recipients is strongly influenced by the source's credibility, which consists of two main aspects: trustworthiness and expertise. As an important information resource, AI's credibility significantly impacts consumers (Diakopoulos 2016). Perceived AI credibility is a core indicator for assessing the reliability and validity of AI (Shin et al. 2020). In the context of AI, trustworthiness is primarily characterized by the fairness, accountability, and transparency of the algorithms. At the same time, expertise is linked to the innovation and industry leadership of the technology (Khan and Mishra 2024). Compared to objective credibility, perceived AI credibility places greater emphasis on consumers' subjective feelings and evaluations, which are influenced by a combination of personal experience, preferences, and cognitive abilities (Shin et al. 2020). Perceived AI credibility represents consumers' comprehensive assessment of artificial intelligence's fairness, transparency, responsibility, and customizability in the context of interactions between AI systems and users (Khan and Mishra 2024).

AISR is closely related to consumers' perceived AI credibility. AI significantly enhances consumer trust, reduces perceived risk, and increases satisfaction (Petrescu et al. 2024). People are generally more inclined to trust AI algorithms (Logg et al. 2019; von Walter et al. 2022). However, when consumers observe an algorithm making a mistake, they are more likely to avoid using it and develop an aversion to it (Dietvorst et al. 2015). Meanwhile, Shin et al. (2020) contend that consumers' objective verification of technological functionality—such as algorithmic explanations—more readily evokes perceptions of trustworthiness than moral commitments. Therefore, only by effectively addressing consumers' concerns about AI technology can companies enhance their trust in AI technology, thereby increasing their loyalty and support for the brand (Wirtz and Lwin 2009). According to source credibility theory, the source's credibility has a strong influence on consumer behavioral decisions. When technical AISR meets individual consumer functional demands, such as protecting customer privacy and eliminating algorithmic bias, consumers perceive the reliability of AI technology, thus establishing perceived AI credibility. Moreover, this increased credibility enables consumers to experience and recognize the company's products more deeply, thereby stimulating their brand favorability (Amos et al. 2008) and purchase intentions (Tan and Liew 2020). Most importantly, as consumers' trust in AI technology grows, they are more likely to become brand loyalists (Kim et al. 2014).

However, according to the Heuristic-Systematic Model (HSM), users employ both heuristic and systematic processes for dual information processing before making behavioral decisions (Chaiken 1980). Heuristic information processing involves individuals making quick judgments through heuristic cues, which do not require excessive cognitive effort. Systematic information processing involves individuals evaluating the information based on rationality and logic, which requires more cognitive resources.

Technical AISR entails functional attributes (e.g., privacy protection, algorithmic fairness) that demand consumers' cognitive resources for rational appraisal. Such AISR—illustrated by algorithmic explainability or data provenance—requires users to verify technological feasibility (e.g., "How does this AI safeguard my privacy?"), aligning with the systematic-processing need for "cognitive effort" (Verplanken and Aarts 1999). Consumers must evaluate technical efficacy through logical analysis (e.g., scrutinizing privacy-policy specifics) rather than relying on affective heuristics (Bonaccio and Dalal 2006). Moreover, consumers' risk-averse tendencies toward technology activate deep information processing, such as verifying data security (Du and Xie 2021). Therefore, Technical AISR (vs. institutional AISR) triggers systematic processing, leading consumers to scrutinize technical attributes (e.g., algorithmic transparency, data security), thereby enhancing perceived AI credibility and brand advocacy. In contrast, consumers make decisions about institutional AISR more from a 'heuristic processing' perspective, making quick judgments without excessive cognitive effort, which is not conducive to establishing perceived AI credibility (Flanagin and Metzger 2007). In summary, the following hypotheses are proposed:

H_2: Perceived AI credibility mediates the effect of AISR on consumer brand advocacy. Compared to institutional AISR, technical AISR is more likely to enhance perceived AI credibility, promoting consumer brand advocacy.

The mediating role of perceived AI empathy. Consumer acceptance of AI depends on the fulfillment of functional needs and emotional fulfillment (Wirtz et al. 2018). AI empathy's emergence represents an important emotional intelligence breakthrough within AI technology (Huang and Rust 2018). AI empathy showcases AI's superior ability to recognize and respond to human cognitive and emotional needs (Asada, 2015), to create an AI system that can empathize with and interact with humans, thereby endowing it with a more humane understanding of emotions (Asada 2015). AI empathy encompasses three dimensions: perspective-taking, emotional concern, and emotional contagion, which steer AI towards greater humanity and intelligence (Liu-Thompson et al. 2022).

With the advancement in AI empathy technology, particularly the development of affective computing, AI entities have become increasingly adept at sensing and responding to consumers' emotional fluctuations (Poria et al. 2017). AI empathy fosters an emotional connection between consumers and brands, which can profoundly impact consumer satisfaction and brand loyalty (Zein 2021). Consumers desire to feel heard, understood, and valued in their interactions with AI. The empathic ability of AI not only enhances consumers' positive impressions of the brand but also significantly influences their purchasing decisions. The positive impacts of various types of human-computer emotional relationships are well-documented, primarily in promoting positive attitudes, increasing the willingness to purchase, and enhancing long-term usage (Liu 2024).

Consumers' information processing of different AISRs directly affects perceived AI empathy, influencing various brand advocacy

behaviors. Institutional AISR (e.g., social welfare, environmental protection) conveys signals of societal values, allowing consumers to simplify their decisions via the "affect heuristic" (Aljarah et al. 2022). Liu-Thompson et al. (2022) demonstrate that appeals to social value are more likely to trigger affect-heuristic responses. Moral appeals readily activate an "emotional shortcut" (e.g., empathic reactions), thereby reducing cognitive costs (Turri et al. 2013). Therefore, institutional AISR(vs technical AISR) triggers heuristic processing, where consumers rely on affective cues (e.g., ethical reputation) for quick judgments, resulting in lower perceived credibility but higher empathy. Therefore, compared to technical AISR, when consumers are exposed to institutional AISR, they form a higher level of empathy towards institutional AISR through the "heuristic processing" of information. In contrast, consumers approach technical AISR through "systematic processing" of information, relying on algorithmic rules to address problems. This approach's processing speed is slower and can avoid various decision-making biases, but it requires more mental resources and effort from decision-makers, which is not conducive to improving perceived AI empathy (Bonaccio and Dalal 2006). In summary, the following hypotheses are proposed:

H_3: Perceived AI empathy mediates the effect of AISR on brand advocacy. Compared with technical AISR, institutional AISR is more likely to enhance consumers' perceived AI empathy, promoting consumers' brand advocacy.

The moderating role of artificial intelligence literacy. AI literacy encompasses a set of competencies that enable individuals to critically evaluate AI technologies, effectively communicate and collaborate with AI, and utilize AI as a powerful tool in online environments, at home, and in the workplace (Long and Magerko 2020). The concept of AI literacy has been interpreted differently by various groups. This paper concentrates on the AI competencies that should be accessible to the general public, particularly for those learners without a computer science background. Thus, in this research context, AI literacy does not delve into the intricacies of programming or algorithms but emphasizes generalized knowledge of AI (Laupichler et al. 2022). With the ongoing progress and widespread popularization of AI technology, there is a growing recognition that, beyond practitioners in the professional field, the general public also needs to possess a certain level of AI literacy to understand, apply, and interact efficiently with AI.

Key components of the AI literacy framework include technical knowledge, ethics, and practical skills (Almatrafi et al. 2024), all of which are closely linked to AISR cognition. Technical knowledge involves understanding fundamental AI concepts, technical principles, application scenarios, and insights into AI technology trends (Long and Magerko 2020). This knowledge forms the foundation for assessing the social impacts of AI technologies; only by understanding how AI operates can the potential social responsibility issues associated with it be anticipated and evaluated (Mo et al. 2024). Ethics involves moral responsibility and behavioral norms in the use of AI technologies. It includes awareness of, and responsiveness to, the potential adverse effects of these technologies, as well as a sense of social responsibility regarding their application (Müller 2021). Ethics lies at the core of AISR, requiring individuals to consider the moral and ethical implications of AI technologies in their use, thereby promoting fair, transparent, and safe AI applications (Müller 2021). Practical skills encompass the ability to use and create AI, focusing on applying AI knowledge, concepts, and tools to solve problems across various scenarios (Ng et al. 2021; Han et al. 2019). These skills empower individuals to effectively harness AI technologies and make responsible decisions in their application, which is essential for achieving AISR. They directly influence the practical application of AI technologies and their positive societal contributions.

According to Leichtmann et al. (2023), moderate trust in AI positively correlates with an individual's AI literacy. Montal and Reich (2017) proposed that algorithmic literacy is related to the credibility of AI news services. Consumers with higher algorithmic literacy attribute more credibility to AI news (Shin 2022). People's level of trust in AI is also largely influenced by the type of task (Long and Magerko 2020). Therefore, a deeper understanding of AI's ability to perform in different task scenarios—consumers with a high level of AI literacy understand AI logic better—can help them make more informed decisions, including how to use AI wisely and when to choose to trust it (Long and Magerko 2020).

While the experience of empathy is an internal mental process, its expression fundamentally occurs within the communicative aspect of human-computer interaction (Shao 2023). An individual's literacy with AI products can shape their mental model (Rosling and Littlemore 2011), and this mental model plays a crucial role in the interaction process (Norman 2013). AI empathy encodes empathy into AI algorithms and intelligences (Asada 2015). Programming empathic care into AI agents involves two fundamental steps. The first is emotion recognition, which entails detecting the consumer's emotional state at various times. Without first recognizing emotions, the AI agent lacks a basis for empathic concern. The second step involves adaptive communication regarding these emotions, a vital and necessary component of empathy (Weißhaar and Huber 2016). On the other hand, AI literacy is a set of abilities that enable individuals to assess AI technology and effectively communicate and collaborate with AI (Long and Magerko 2020). Therefore, consumers with high AI literacy can critically evaluate AI technologies, effectively communicate and collaborate with AI, and understand and embody the empathic qualities reflected in institutional AISR. Consequently, they are more likely to develop a stronger affinity for identifying with and advocating for the brand.

In summary, AI literacy is positively correlated with consumers' moderate trust in AI; that is, the higher the AI literacy, the greater the consumers' capacity to comprehend and assess AI technologies, leading to higher perceived AI credibility towards technical AISR. Concurrently, AI literacy also positively influences consumers' perceived AI empathy for institutional AISR. Thus, AI literacy is relevant to consumers' perceptions and trust in AI technology. However, it also plays an essential moderating role in shaping consumers' psychological experiences and brand attitudes during their interactions with AI. Based on this, the following hypothesis is proposed:

H_4: AI literacy moderates the role of AISR in influencing brand advocacy, and this role is mediated through perceived AI credibility. That is, the higher the AI literacy, the higher the perceived AI credibility of consumers towards technical AISR relative to low AI literacy.

H_5: AI literacy moderates the role of AISR in influencing brand advocacy, and this role is mediated through perceived AI empathy. That is, the higher the AI literacy, the higher the perceived AI empathy of consumers towards public AISR relative to low AI literacy.

In summary, the research model presented in this paper is derived and depicted in Fig. 3.

Methodology

All experiments conducted in this study (May-July 2024) utilized G*Power 3.1 software for a priori statistical power analysis to determine the required sample size. For Experiment 1, the F-test [ANOVA: Fixed effects, omnibus, one-way] ANOVA method was

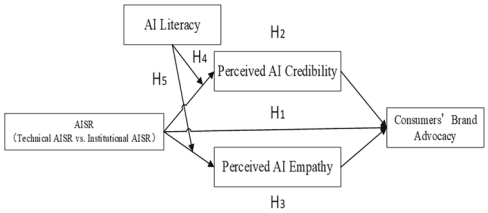

Fig. 3 Research model.

Table 4 Sample size required for prior statistical power analysis.

Experiment	Power	Pre-experiment		Formal experiment	
		Effect size	Sample	Effect size	Sample
study 1	0.9	0.5	45	0.25	172
study 2	0.9	0.5	45	0.25	231

chosen. The pre-tests for Experiment 1 and Study 1 were designed to test for an efficacy power (1-β) of 0.9 and an effect size of 0.5. According to Cohen (Cohen 2013), an effect size of 0.5 represents the threshold for a significant effect and is considered appropriate for experimental pre-tests and manipulation checks. The formal experiments were conducted with an efficacy power (1-β) of 0.9 and an effect size of 0.25. An effect size of 0.25 is considered a medium effect, reflecting that the actual use of service AI is subject to uncontrollable factors that can reduce the effect size. For Study 2, the F-test [ANOVA: Fixed effects, special, main effects, and interactions] analysis of variance method was selected. The test power (1-β) was set to 0.9, the generalized minimum level of efficacy, for both the pre-test and the formal experiment. The pre-test was set with an effect size of 0.5, and the formal experiment was set with an effect size of 0.25, involving four groups and 3 degrees of freedom. See Table 4.

Study1: Mediating effects of perceived AI credibility and AI empathy. The primary objective of Study 1 was to assess the impact of AISR—comparing technical AISR and institutional AISR—on consumer brand advocacy and investigate the mediating roles of perceived AI credibility and perceived AI empathy, thereby validating Hypotheses 1, 2, and 3.

Pre-experiment. Before conducting the formal experiment, a pre-test was carried out to validate the experimental materials. In the pre-experiment, a combination of text and images was utilized to set up the contextual experiment, following the approach of Aaker et al. (2004). Participants were instructed to carefully read the textual and pictorial materials to understand the experimental scenario and then complete the corresponding tests. Concurrently, to differentiate the various dimensions of AISR, the introduction and linguistic conventions were manipulated in this study. Please read the experimental material titled 'Background of the Training Institution (A technical training center that specifically offers education and certification programs related to artificial intelligence for professionals from all walks of life),' which consists of text and images, and then complete the related experimental questionnaire. We standardized image style, resolution, and layout across all conditions to control for potential confounding effects of visual or emotional cues in the manipulation materials. All images used neutral color palettes and contained no emotionally arousing elements (e.g., affective gestures, warm/cool color tones). Text descriptions were also rigorously matched across conditions regarding character count (80 ± 5 characters), syntactic complexity (Flesch–Kincaid readability score), and neutral tone. Only the manipulated variable—the type of AISR (technical vs. institutional)—was varied, while narrative structure and expressive style remained identical across groups. A pilot test further confirmed that participants did not perceive unintended differences in emotional tone or vividness unrelated to the AISR manipulation. Results showed no significant difference between the two versions in perceived emotional intensity ($p > 0.10$), validating that the manipulation was emotionally neutral and visually comparable. The detailed experimental materials can be found in Appendix 2.

Forty-six participants were recruited on the Credamo platform and randomly assigned to two groups to review the material. Subsequently, they were asked questions about AISR. A 7-point Likert scale was utilized, with one indicating 'strongly disagree' and 7 indicating 'strongly agree'. Paired-samples t tests revealed significant differences between the experimental groups: for the technical AISR group, $M_{\text{technical}} = 6.125$, $M_{\text{institutional}} = 3.895$, $t = 7.031$, $p < 0.01$; for the institutional AISR group, $M_{\text{technical}} = 4.909$, $M_{\text{institutional}} = 5.818$, $t = -2.969$, $p < 0.05$. Therefore, the experimental materials satisfied the manipulation requirements, allowing for further formal experimental studies.

Formal experiment. Formal Experiment 1 included measures of perceived AI credibility, perceived AI empathy, and brand advocacy. Perceived AI credibility was assessed using the scale from Shin et al. (2022) and Masuda et al. (2022), which comprises four question items. Perceived AI empathy was evaluated using the scale from Liu-Thompkins et al. (2022), consisting of three question items. Brand advocacy was assessed using the framework from Melancon et al. (2011), while AISR was evaluated using the scale developed in this study. All items were rated on a 7-point Likert scale, where 1 indicated 'strongly disagree' and 7 indicated 'strongly agree'. Additionally, gender, age, education, occupation, and years of AI use served as control variables.

Study 1 recruited 330 participants through the Credamo platform and randomly assigned them to two experimental groups. Twenty-two invalid questionnaires were excluded, resulting in 308 valid questionnaires retained, with a questionnaire validity rate of 93.3%. Among the participants, 115 were male and 193 were female; 67% held bachelor's degree, and 57% were 25-35. Paired-samples t tests revealed significant differences between the experimental groups (for technical AISR: $M_{\text{technical}} = 5.880$, $M_{\text{institutional}} = 3.301$, $t = 21.368$, $p < 0.001$; for institutional AISR: $M_{\text{technical}} = 4.623$, $M_{\text{institutional}} = 5.928$, $t = -8.091$, $p < 0.001$), indicating that the experimental materials successfully manipulated the independent variables. Additionally, the Cronbach's α values for perceived AI credibility, AI empathy, and brand advocacy ranged from 0.743 to 0.803, all exceeding the criterion of 0.700.

Results. Based on the one-way ANOVA test, it was found that perceived AI credibility significantly differed between the two experimental groups($M_{\text{technical}} = 5.883$, $M_{\text{institutional}} = 5.161$, $F(1, 306) = 50.244$, $p < 0.001$). The technical AISR group scored higher than the institutional AISR group, thus increasing perceived AI credibility. Perceived AI empathy was higher in the institutional AISR group compared to the technical AISR group ($M_{\text{technical}} = 3.749$, $M_{\text{institutional}} = 5.20$, $F(1, 306) = 502.284$, $p < 0.001$). Consistent with the results of Study 1, Consumer brand advocacy significantly differed between the two experimental groups ($M_{\text{technical}} = 5.404$, $M_{\text{institutional}} = 5.841$, $F(1, 306) = 30.775$, $p < 0.001$), confirming Hypothesis 1. Additionally, we conducted an ANCOVA, treating AISR type (technical vs. institutional) as the independent variable and brand advocacy as the dependent variable, while controlling for relevant

Table 5 Mediating effects of perceived AI credibility and AI empathy.

	Mediating variable	Effect	SE	t	p	95%	
						LLCI	ULCI
Direct Effect	-	0.2100	0.1281	1.6399	0.1021	−0.0420	0.4620
Mediating Effect	AI Credibility	-0.0941	0.0250	-	-	−0.1429	−0.0458
	AI Empathy	0.3208	0.1090	-	-	0.1163	0.5458

Table 6 Mediating effects of perceived AI credibility and AI empathy in Study 2.

	Mediating variable	Effect	SE	t	p	95%	
						LLCI	ULCI
Direct Effect	-	0.1183	0.1302	-0.9087	0.3643	−0.1397	0.3745
Mediating effect	AI Credibility	-0.3255	0.0621	-	-	−0.4607	−0.2137
	AI Empathy	0.3305	0.0886	-	-	0.1687	0.5132

demographic and AI experience variables (gender, age group, education, occupation, and years of AI usage). Results showed that the effect of AISR type on brand advocacy remains statistically significant even after controlling for these covariates ($F = 16.48$, $p < 0.001$). This confirms that the observed group differences are not attributable to background variables.

Next, this study treated AISR as a categorical variable, coding technical AISR as zero and institutional AISR as 1. Following Zhao et al.'s (2010) proposed moderated mediation effects analysis model (Model 4), Bootstrap sampling with 5,000 replications was selected to test further the mediating effects of perceived AI credibility and AI empathy. The mediation analysis results, as shown in Table 5, indicated that the direct effect of AISR on brand advocacy was not significant (95% CI: [−0.0420, 0.4620], with zero included). The indirect effects of perceived AI credibility (95% CI: [−0.1429, −0.0458], interval not containing 0) and perceived AI empathy (95% CI: [0.1163, 0.5463], interval not containing 0) were significant. Thus, AISR influences consumer brand advocacy through the mediating roles of perceived AI credibility and AI empathy. Technical AISR can enhance perceived AI credibility more than institutional AISR, which in turn promotes consumer brand advocacy. Conversely, institutional AISR can enhance perceived AI empathy more than technical AISR, which also promotes consumer brand advocacy. Therefore, Hypotheses 2 and 3 were validated.

Study 2: Moderating effects of AI literacy. Study 2 employed a 2 (AISR: technical AISR vs. institutional AISR) × 2 (AI literacy: high vs. low) between-groups experimental design to test the moderating effect of AI literacy, thereby examining Hypotheses 4 and 5.

Pre-experiment. Select the experimental material, including text and images detailing the 'e-commerce platform background.' The reason for choosing the background experimental materials of e-commerce is that they represent the most common field of AI interaction (such as recommendation, personalization, and automated services). Thus, it is suitable for testing the influence of individual-level regulators such as AI literacy. After reviewing all the experimental materials, participants completed the relevant experimental questions. Detailed experimental materials are presented in Appendix 3.

Measurement variables included the existing variables from Study 1 and AI literacy (AIL) variables. Referring to Wang et al.'s (2023) study, the AIL scale comprises 4 question items. Forty-eight participants were randomly assigned to two experimental groups for this pre-experiment; the final number of valid samples was 45. The paired-samples t test indicated significant differences in AISR between the two experimental groups: for the Technical AISR group, $M_{\text{technical}} = 5.729$, $M_{\text{institutional}} = 3.896$, $t = 6.196$, $p < 0.01$; for the Institutional AISR group, $M_{\text{technical}} = 4.810$, $M_{\text{institutional}} = 5.809$, $t = -2.291$, $p < 0.05$. Therefore, the experimental material satisfies the manipulation requirements, allowing for further formal experimental studies.

Formal experiment. The formal experiment recruited 320 participants through the Credamo platform. After excluding 20 invalid questionnaires, 300 valid questionnaires were retained, resulting in a validity rate of 93.7%. Among the participants, 105 were male and 195 were female. The age distribution was primarily between 26 and 35 years old, accounting for 57.3%, and the proportion with a bachelor's degree was 77.3%. Paired-samples t tests revealed significant differences between the experimental groups: for the technical AISR group, $M_{\text{technical}} = 5.876$, $M_{\text{institutional}} = 4.037$, $t = 14.495$, $p < 0.001$; for the institutional AISR group, $M_{\text{technical}} = 4.487$, $M_{\text{institutional}} = 5.826$, $t = -9.609$, $p < 0.001$. These results indicate that the experimental material successfully manipulated the independent variable. Additionally, the Cronbach's α values for AI literacy, perceived AI credibility, perceived AI empathy, and brand advocacy ranged from 0.729 to 0.852, all exceeding the threshold of 0.700.

Results. In this study, AISR was converted into a dummy variable, with '0' representing technical AISR and '1' representing institutional AISR. The sample sizes for the technical AISR experimental group and the institutional AISR experimental group were both 150. Using a one-way ANOVA, it was found that technical AISR resulted in higher perceived AI credibility than institutional AISR ($M_{\text{technical}} = 5.783$, $M_{\text{institutional}} = 4.890$, $F(1, 298) = 64.914$, $p < 0.001$). In contrast, institutional AISR resulted in higher perceived AI empathy than technical AISR ($M_{\text{technical}} = 4.042$, $M_{\text{institutional}} = 5.700$, $F(1, 298) = 151.699$, $p < 0.001$). Subsequently, Model 4 in the SPSS PROCESS program was employed to set the Bootstrap count to 5000 (Zhao et al. 2010) to test the mediating effects of perceived AI credibility and perceived AI empathy. The mediation analysis indicated that the direct effect of AISR on brand advocacy was not significant (95% CI: [−0.1379, 0.3745], inclusive of 0), whereas the indirect effects of perceived AI credibility (95% CI: [−0.4607, −0.2137], interval not containing 0) and perceived AI empathy (95% CI: [0.1687, 0.5132], interval not containing 0) reached significant levels. The results are presented in Table 6. Therefore, AISR can influence consumer brand advocacy through the dual mediating effects of perceived AI credibility and AI empathy, and Hypotheses 2 and 3 are again confirmed.

Table 7 AI literacy moderating role.

Source	DV	III SS	df	MS	F	P
AISR * AI literacy	AI credibility	65.158	3	32.579	35.908	<0.001
	AI empathy	204.154	3	103.077	75.615	<0.001

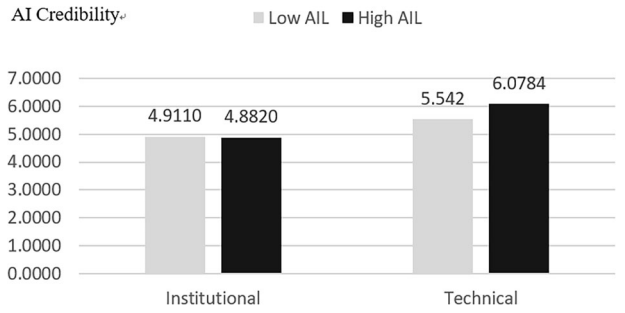

Fig. 4 Moderating effect of AI literacy on AISR and perceived AI credibility.

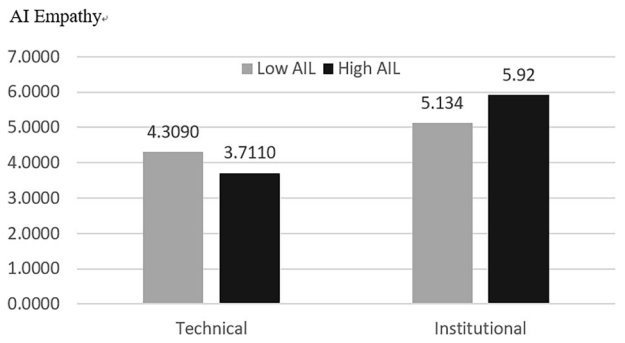

Fig. 5 Moderating effect of AI literacy on AISR and perceived AI empathy.

AI literacy was measured using a 7-point Likert scale, which categorized the continuous variable of AI literacy for analysis of variance. The mean of the AI literacy scores (Mean = 5.6267) served as the basis for classification: participants with AI literacy scores below the mean were designated as the low AI literacy group, coded as '0', and those with scores above the mean were designated as the high AI literacy group, coded as '1'. In this study, a group with a high AI literacy score is represented by '1'. Subsequently, a two-way ANOVA with perceived AI credibility as the dependent variable revealed a significant interaction between AISR and AI literacy ($F = 35.908$, $p < 0.001$). This suggests that the effect of AISR on perceived AI credibility is moderated by AI literacy, as shown in Table 7.

Specifically, higher AI literacy is associated with greater perceived AI credibility from technical AISR compared to lower AI literacy ($M_0 = 5.5452 < M_1 = 6.0784$), as illustrated in Fig. 4. Thus, Hypothesis 4 is validated. Similarly, a two-way ANOVA conducted in this study with perceived AI empathy as the dependent variable revealed a significant interaction between institutional AISR and AI literacy on perceived AI empathy ($F = 76.615$, $p < 0.001$), indicating that the effect of institutional AISR on perceived AI empathy is moderated by AI literacy, as shown in Table 7. Specifically, higher AI literacy is associated with greater perceived AI empathy for institutional AISR than lower AI literacy ($M_0 = 5.134 < M_1 = 5.920$), as seen in Fig. 5. Thus, Hypothesis 5 is validated.

Discussion

Results. This paper examines the dimensional structure of AISR, develops a corresponding measurement scale, and explores the mechanisms and boundary conditions of its influence on consumer brand advocacy.

First, using grounded theory, this paper derived a two-dimensional structure of AISR, consisting of technical AISR and institutional AISR, along with the initial measurement items. Specifically, technical AISR encompasses three additional factors: safety, usability, and reflexivity, while institutional AISR includes two additional factors: ethics and well-being. Building on this foundation, the paper further developed an AISR measurement scale comprising 18 items, utilizing exploratory factor analysis, confirmatory factor analysis, and structural validity analysis based on questionnaire data.

Secondly, drawing on the Heuristic-Systematic Model (HSM), this paper further develops a model and research hypotheses regarding the impact of AISR on consumer brand advocacy and conducts two experimental studies. Study 1 indicates that institutional AISR is more likely to stimulate consumer brand advocacy than technical AISR. This suggests that consumers' perceptions of AI technology's social responsibility extend beyond the safety and reliability of the technology itself to include its impact on society, the environment, and ethics. While previous literature has examined the impact on consumer behavior in terms of CSR corporate service motivation and public service motivation (Aljarah et al. 2022), or analyzed the impact from a single dimension of AISR (Castro-González et al. 2019), this paper provides new insights into the differentiation mechanism of consumer brand advocacy behavior within the AISR framework. Study 1 also shows that perceived AI credibility and perceived AI empathy mediate the relationship between AISR and consumer brand advocacy. As hypothesized, technical AISR enhanced perceived AI credibility to a greater extent than institutional AISR, thereby promoting brand advocacy. Conversely, institutional AISR enhanced perceived AI empathy more than technical AISR, which in turn fostered brand advocacy. Study 2 demonstrates that AI literacy positively moderates the relationship between AISR effects and perceived AI credibility, as well as AI empathy. Specifically, consumers with higher AI literacy reported higher perceived AI credibility in response to technical AISR, and higher perceived AI empathy in response to institutional AISR, compared to those with lower AI literacy.

Theoretical implications. This paper proposes and empirically tests a two-dimensional structure of AISR (technical AISR and institutional AISR) using a grounded theory approach in the field of consumer research. It extends traditional CSR theories, such as Carroll's CSR pyramid and stakeholder theory, to the context of AI, and explores new dimensions, including "reflexivity." This work enriches the theoretical understanding of AISR. We demonstrate how technical and institutional AISR map onto the distinct layers of corporate responsibility and stakeholder networks, thereby providing a framework that can facilitate future research on AI ethics and responsibility. This paper also helps to clarify the conceptualization and dimensional structure of AISR in the consumer domain, contributing to the evolving theoretical framework of AI-driven social responsibility. As AISR research is still in its preliminary stage, there is no consensus within the academic community regarding the conceptual connotation and dimensional structure of AISR. Against this backdrop, this paper defines the concept of AISR in the context of consumption and elucidates its nature within the consumer sphere. Furthermore, this study not only builds on and extends the general commonality dimensions identified in previous research, such as ethics (Kumar et al. 2023), safety (Ratzan and Rahman 2023), and privacy protection (Martin and Zimmermann 2024), but also explores reflexivity as a dimension of AISR. It reveals the

multifaceted composite structure of AISR in the consumer domain. The conceptual and dimensional clarity offered by this paper provides a basis for subsequent empirical research in this area.

Secondly, the measurement tool introduced in this paper is innovative. Most existing research relies on single-dimension or context-specific scales. In contrast, this study develops and validates a dual-dimensional, five-factor AISR scale consisting of 18 items, which demonstrates strong reliability and validity. This provides a standardized measurement foundation for future empirical research on AISR. Currently, some studies' measurements of AISR are primarily derived from the research results of responsible AI (Kumar et al. 2023), which reduces the systematicity, relevance, and scientific validity of the studies. Therefore, the AISR scale developed in this study within the field of consumption offers a reliable measurement tool for subsequent quantitative research.

Thirdly, this paper further enriches the understanding of AISR in the consumer domain from the consumer-brand relationship perspective, clarifying the mechanisms and boundary conditions through which AISR influences consumer brand advocacy. While most existing studies broadly suggest that social responsibility enhances customers' consumption experiences (Du and Xie 2021), this paper more precisely identifies the two pathways of AISR's influence on consumer brand advocacy in the consumer domain: perceived AI credibility and AI empathy. Compared with studies on how CSR affects brand relationships—which tend to emphasize the positive impact of ethical responsibility—our findings show that, in the AI context, technical responsibility can likewise enhance consumer brand advocacy. This effect is mediated by AI trustworthiness. Moreover, whereas traditional digital-ethics literature centers on "minimum obligations" such as data privacy and transparency, this study reveals that "ethics-by-design" and "emotional care" within institutional AISR have become key sources of consumers' active brand advocacy. This finding hints at a potential shift in consumers' perception of digital ethics, suggesting it may be evolving from a "defensive morality" (focused on minimum obligations) toward a "relational morality" (emphasizing emotional care and ethical design). Furthermore, this paper validates the moderating role of AI literacy in the aforementioned mediating mechanisms. The findings of this study enrich and expand the post hoc research on AISR in the consumer domain.

Practical implications. First, enterprises should place great emphasis on fulfilling AISR. Companies can strategically incorporate institutional AISR themes into their marketing narratives, positioning AI ethics and governance as part of their brand identity. Phrases such as "AI that cares," "Responsible Intelligence," or "Built for fairness" can enhance emotional bonding and advocacy. Responsible practices can contribute to enhancing a brand's image and garnering consumer support as companies apply AI technology. In the long run, this approach leads to the company's sustained success and aligns with broader societal interests, creating a harmonious win–win outcome.

Secondly, companies should endeavor to enhance consumers' perception of trust and empathy towards AI technologies. To bolster consumer trust, firms should increase the transparency and accountability of AI technologies, ensuring that algorithmic decisions are fair and reasonable. Twitter's META team, for instance, is primarily responsible for assessing potential unintentional harms caused by algorithms and prioritizing pending issues (Lopez 2023). Concurrently, Twitter has organized the Algorithm Bounty Challenge, employing a community-led approach to develop better algorithms by gathering feedback from diverse people (Lopez 2023). This approach involves communicating the benefits of AI use to consumers and enhancing transparency and control over the use of personal data. Firms could consider embedding institutional AISR cues in user-facing AI interfaces, for instance, by disclosing oversight by an ethics board or adherence to specific governance standards. Labels such as "Ethically Audited" or "Human-in-the-Loop" have the potential to strengthen user trust and empathy toward the system, as indicated by our findings on perceived AI empathy. Additionally, companies should identify and respond to customers' emotions by refining AI sentiment analysis technology. This enables more personalized and humanized services, thereby enhancing users' emotional experiences and fostering brand advocacy.

Finally, companies should adopt a tiered communication strategy when communicating their AISR to consumers with varying levels of AI literacy. Companies should offer verifiable transparency features for users with high AI literacy, such as algorithmic audit trails, ethical compliance dashboards, or model cards. These build cognitive trust and align with their technical clarity and preference for accountability. For low-literacy audiences, highlight emotional safety, empathy, and human-centered values in the AISR messaging.

Limitations and future research. First, the generalizability of our findings is constrained by the sample, which was drawn exclusively from a single Chinese online platform (Credamo). The perception and weight of AISR dimensions, as well as the level of AI literacy, may vary significantly across cultures (e.g., individualistic vs. collectivistic societies) and regulatory environments (e.g., regions with strict AI laws like the EU vs. others). To address this, future research should validate this model in diverse cultural and demographic settings. Furthermore, sensitivity analyses should be conducted to determine the minimum detectable effect sizes given the achieved sample sizes and desired power level. These additional tests will enhance the accuracy of understanding the performance of the measurement tools and provide a more robust foundation for conducting causal mediation analysis. Additionally, it is essential to note that our experimental scenarios (training institution and e-commerce platform), although common, represent only a subset of AI service contexts. The salience of technical vs. institutional AISR might differ in other domains, such as healthcare, finance, or autonomous vehicles, where risks and ethical concerns are more pronounced. Therefore, examining the boundary conditions of our effects across different service types represents a valuable direction for future studies.

Second, given the focus on ethical and socially responsible behavior, the study may be susceptible to social desirability bias in self-reported measures. Future research could employ indirect questioning techniques or behavioral measures to mitigate this potential bias. It is crucial to address this potential bias to enhance the validity of the findings, particularly for constructs related to ethics and social responsibility. To mitigate and account for social desirability bias, consider the following strategies: Incorporate a validated social desirability scale, such as the Marlowe-Crowne Social Desirability Scale, and report its correlations with key study variables; Implement indirect questioning techniques for sensitive items, such as the randomized response technique or the unmatched count technique; Utilize scenario-based measures that reduce the personal nature of responses and potentially decrease socially desirable responses; Employ a mix of positively and negatively worded items to detect acquiescence bias.

Third, the dichotomous classification of AISR into technical and institutional dimensions, while theoretically sound, may oversimplify reality. Future research could employ methods such as latent profile analysis to identify more nuanced combinations of AISR attributes. Real-world implementations of AISR often involve hybrid attributes (e.g., privacy-preserving algorithms that also promote fairness), which may defy neat classification. Future research should explore dynamic configurations of AISR using more flexible analytic approaches such as latent profile analysis or fuzzy-set qualitative comparative analysis (fsQCA) to identify nuanced combinations of AISR practices and their effects.

Finally, this study examined two specific mediators (perceived AI credibility and empathy) and one moderator (AI literacy). Whether other unexamined mechanisms, such as perceived justice or moral identity, could also explain the relationship between AISR and brand advocacy presents a valuable avenue for future research. In the experimental study, despite the non-significant direct effect, this may suggest that the indirect effects via the mediators of AI credibility and AI empathy constitute a "primary" or "dominant" pathway through which AISR influences brand advocacy, rather than the sole route. Research could therefore continue to explore additional, potentially secondary pathways.

Data availability

Replication data for the paper are available at https://figshare.com/s/9446c3b191e3ebeccca0.

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Author contributions

Pengyi Shen: Research design, theoretical framework; Dehui Peng: Manuscript drafting, literature review, data analysis, and interpretation. Aonan Zhu: Guidance on analytical methods. Jinan Xu: Research design, revision guidance. After contributing to the manuscript, all four authors have approved the submission of this paper.

Competing interests

The authors declare no competing interests.

Ethical approval

The survey conducted in this study adhered to the standards outlined in the Declaration of Helsinki. The Jiangxi University of Finance and Economics Institutional Review Board granted ethical approval. Approval number: 202401-MK-0830-01. Date of approval: 01.01.2024, Scope of approval: All research variables in this study.

Informed consent

Written informed consent was obtained from the participants by the corresponding author of this paper from January 1 to June 30, 2024. The data for this empirical research were generated through a survey, during which the questionnaire was administered. All respondents to the survey questionnaire were asked to complete it online. The participants' participation was voluntary. The background of the study, its purpose, proposed activities, potential benefits and outcomes, possible risks, and the confidentiality of the

records were explained. The participants were informed that they were free to withdraw from the survey at any point in time. Before participation, all participants were informed about the aim of the research and assured of the anonymity of their data. After providing informed consent for the study, the participants duly signed the form, and the questionnaire was activated. The above statements were approved by the School of Business Administration at Jiangxi University of Finance and Economics in July 1, 2024.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1057/s41599-025-06344-z.

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